CV

AI CAD-to-Cost Analysis

CostVision · Powered by Claude AI · Manufacturing Intelligence Platform

AI-POWERED CAD ANALYSIS REPORT

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Geometry to Process to Cost · Fully AI-reasoned

18

PRCR002 – Forged Structural/Drivetrain Component

277 × 223 × 182 mm · 1037.1 cm³ · Al 2.800 kg / Steel 8.141 kg

Manufacturability: 18/100 · Confidence: Low

§1 — Geometry & Part Summary

Bounding Box	**276.8 × 223.2 × 181.9 mm**	Surface Area	**1370.1 cm²**
Volume	**1037.11 cm³**	Weight (Al)	**2.800 kg**
Weight (Steel)	**8.141 kg**	Weight (Plastic)	**1.089 kg**

§2 — Detected Features

Feature Type	Count	Significance	Description
Holes (drilled/bored)	21	**High**	21 holes r<30mm requiring post-forge drilling/boring/reaming
Threaded features	1	**Medium**	Tapped/threaded bores detected — cannot be forged net-shape, machined after forging
Boss/shaft features	7	**High**	Large diameter bosses/shafts (r=10–57mm) forming main load-bearing sections
Undercut faces	44	**High**	44 faces with negative draft relative to Z draw direction — conflicts with single-plane forging draw
Free-form (B-spline) surfaces	76	**Medium**	76 organic/blended surfaces increasing die-sinking complexity

§3 — Material Analysis

AI-suggested

Primary Material: **38MnVS6 Microalloyed Forging Steel (62% confidence)**

Thick, dense, complex solid (fill ratio 0.092 driven by internal geometry not thin shell) with bosses/shafts up to 57mm radius, multiple bores and threads — classic profile of a forged automotive drivetrain/suspension component (knuckle/arm/hub-type part). Per guidance, thick-section forged parts of this nature default to steel rather than aluminium absent an explicit lightweighting signal. 38MnVS6 is a precipitation-hardening microalloyed steel purpose-built for hot closed-die forging without post-forge heat treatment, common in this part class.

Alternatives: **AISI 4340 Alloy Steel (22%)** · **AISI 4130 Alloy Steel (12%)** · **Al 7075 Forging Alloy (4%)**

§4 — Process Recommendations

Process	Commodity	Confidence	Cycle Time (hr)	Reasoning
Closed-die hot forging	forging	95%	0.0300	User-forced ground truth; part geometry (thick bosses, moderate free-form blending, high strength requirement implied by drivetrain-type shape) suits closed-die forging at 200k/yr volume, followed by machining of holes/threads.
CNC finish machining (post-forge)	machining	60%	0.6699	21 holes, threaded features, and 3 machining setups are required after forging to achieve final bores/threads; bottom-up CNC estimate used verbatim.

§5 — Manufacturability Risks (Score: 18/100)

Severity	Feature / Area	Description	Recommended Action
High	**Undercuts (44 faces)**	Numerous undercut faces relative to single draw direction cannot be forged in one blow; will require die re-orientation, split-die design, or secondary machining to form these features.	Re-orient parting line to minimize undercuts or convert undercut features to post-forge machined details.
High	**Non-uniform section (Ã = 4 6 . 6 m m)**	Very high variance in wall/section thickness (0.45mm to 170mm) risks incomplete die fill, cold shuts, and flow-line discontinuities in thin regions adjacent to massive bosses.	Add fillets/ribs to smooth section transitions and add draft to ease metal flow into thin zones.
Medium	**Threaded/drilled features**	21 holes plus threads cannot be forged net-shape and require a dedicated multi-setup machining line, adding cost and cycle time at 200k/yr volume.	Consolidate hole operations on one setup where possible; consider coining pilot holes in the forging to reduce drilling stock removal.
Medium	**Free-form surfaces (76 faces)**	High B-spline face content increases die-sinking (CNC) time and inspection complexity for the forging die cavity.	Simplify blended surfaces to standard radii/fillets where functionally permissible to reduce die cost.

§6 — DFM Issues (forging)

Severity	Area	Description	Impact	Fix
High	**Undercuts / draw direction**	44 undercut faces conflict with a single Z-axis draw direction typical of closed-die forging.	Requires split dies, side-relief inserts, or extra machining — increases tooling cost by 15–25% and lead time.	Reorient part or split the die along a plane that eliminates most undercuts; move remaining undercuts to post-forge machining.
High	**Large boss/shaft transitions**	Boss radii up to 57mm adjacent to sections down to 0.45mm nominal create abrupt metal-flow gradients in the die cavity.	Risk of laps, cold shuts, and incomplete fill leading to scrap rates >5% at start-of-run.	Increase fillet radii at boss-to-web transitions and add g e n e r o u s d r a f t ("e 7 °) t o a i d me flow.
Medium	**Threaded & precision bores (21 holes)**	Threads and close-tolerance bores cannot be forged net-shape and require full secondary machining across 3 setups.	Adds ~0.67hr machining + 0.75hr setup per part, dominating unit cost at high volume.	Pre-form pilot holes in the forging to reduce stock removal; fixture design to combine setups.
Medium	**Free-form die cavity (76 B-spline faces)**	High proportion of blended/organic surfaces increases 5-axis die-sinking time and die polishing cost.	Die cost and lead time increase 10–20% versus a simpler faceted geometry.	Simplify non-functional blended surfaces to standard radii/fillets to reduce die-sinking time and polishing effort.

§7 — Cost Range & Suggested Inputs

OPTIMISTIC	MOST LIKELY	CONSERVATIVE
£0.00	£0.00	£0.00

Net Weight	**8.141 kg**	Material	**steel**
Recommended Process	**forging**	Cycle Time	**0.6699 hr/part**
Setup Time	**0.750 hr**	Operations	**3 ops**
Operations Detail	**CNC Milling Setup 1 (+Z, 73 faces) (mach-vmc5, 0.3000 hr) CNC Milling Setup 2 (+Y, 71 faces) (mach-haas-vf2, 0.1950 hr) Drilling/Boring 21 holes + tap (mach-drill, 0.1750 hr)**		

Process-Specific Parameters

Casting Subtype	**sand**
Die/Mould Cost	**£0**
Die Life	**0 shots**
Cavities	**0**
Yield	**0.0%**
Flash Weight	**0.814 kg**
Yield	**74.3%**
Die Cost	**£36,141**
Strokes	**8**
Cavities	**0**
Wall Thickness	**0 mm**
Mould Cost	**£0**
Mould Life	**0 shots**
Projected Area	**187.6 cm²**

§8 — AI Analysis Explanation